DEPLOYMENT

CLONE GITHUB

```
eee_W_5998079@runweb219096:~$ aws sts get-caller-identity --query "Arn" --output text
arn:aws:sts::211125455591:assumed-role/voclabs/user4819398=Abidemi_Zainab_Akerejah
eee_W_5998079@runweb219096:~$ aws configure get region
us-east-1
eee_W_5998079@runweb219096:~$ cd ~
eee_W_5998079@runweb219096:~$ git clone https://github.com/icdfa/GRC208-AWS-Capstone-Project.git
Cloning into 'GRC208-AWS-Capstone-Project'...
remote: Enumerating objects: 60, done.
remote: Counting objects: 100% (60/60), done.
remote: Compressing objects: 100% (48/48), done.
remote: Total 60 (delta 16), reused 56 (delta 12), pack-reused 0 (from 0)
Unpacking objects: 100% (60/60), done.
Checking connectivity... done.
eee_W_5998079@runweb219096:~$
```

```
eee_W_5998079@runweb219096:~$ cd GRC208-AWS-Capstone-Project
eee_W_5998079@runweb219096:~/GRC208-AWS-Capstone-Project$ ls -la
total 276
drwxrwxrwx 4 eee_W_5998079 eee_S_3193412  6144 Mar 29 16:54 .
drwxrwx--- 6 eee_W_5998079 apache         6144 Mar 29 16:54 ..
-rw-rw-rw- 1 eee_W_5998079 eee_S_3193412  3551 Mar 29 16:54 architecture_design.md
-rw-rw-rw- 1 eee_W_5998079 eee_S_3193412  2183 Mar 29 16:54 architecture-diagram.md
-rw-rw-rw- 1 eee_W_5998079 eee_S_3193412 13187 Mar 29 16:54 AWS_SERVICES_GUIDE.md
-rw-rw-rw- 1 eee_W_5998079 eee_S_3193412 11185 Mar 29 16:54 BEST_PRACTICES.md
-rw-rw-rw- 1 eee_W_5998079 eee_S_3193412  8810 Mar 29 16:54 cloudformation-database-stack.yaml
-rw-rw-rw- 1 eee_W_5998079 eee_S_3193412  6959 Mar 29 16:54 cloudformation-network-stack.yaml
-rw-rw-rw- 1 eee_W_5998079 eee_S_3193412 11238 Mar 29 16:54 DELIVERY_SUMMARY.md
-rw-rw-rw- 1 eee_W_5998079 eee_S_3193412 11945 Mar 29 16:54 DEPLOYMENT_GUIDE.md
-rwxrwxrwx 1 eee_W_5998079 eee_S_3193412  6985 Mar 29 16:54 deploy.sh
drwxrwxrwx 2 eee_W_5998079 eee_S_3193412  6144 Mar 29 16:54 diagrams
drwxrwxrwx 8 eee_W_5998079 eee_S_3193412  6144 Mar 29 16:54 .git
-rw-rw-rw- 1 eee_W_5998079 eee_S_3193412   545 Mar 29 16:54 .gitignore
-rw-rw-rw- 1 eee_W_5998079 eee_S_3193412 10995 Mar 29 16:54 grc-dashboard.css
-rw-rw-rw- 1 eee_W_5998079 eee_S_3193412 11979 Mar 29 16:54 grc-dashboard.jsx
-rw-rw-rw- 1 eee_W_5998079 eee_S_3193412  9996 Mar 29 16:54 lambda_compliance_monitor.py
-rw-rw-rw- 1 eee_W_5998079 eee_S_3193412 15630 Mar 29 16:54 PRESENTATION_GUIDE.md
-rw-rw-rw- 1 eee_W_5998079 eee_S_3193412 48935 Mar 29 16:54 PRESENTATION_SCRIPT.md
-rw-rw-rw- 1 eee_W_5998079 eee_S_3193412 10971 Mar 29 16:54 PROJECT_MANIFEST.md
-rw-rw-rw- 1 eee_W_5998079 eee_S_3193412 21421 Mar 29 16:54 README.md
-rw-rw-rw- 1 eee_W_5998079 eee_S_3193412   175 Mar 29 16:54 requirements.txt
-rw-rw-rw- 1 eee_W_5998079 eee_S_3193412 10489 Mar 29 16:54 sample_data.sql
-rw-rw-rw- 1 eee_W_5998079 eee_S_3193412 13656 Mar 29 16:54 test_cases.py
eee_W_5998079@runweb219096:~/GRC208-AWS-Capstone-Project$
```

Phase 1: Network Infrastructure

```
eee_W_5998079@runweb219096:~/GRC208-AWS-Capstone-Project$ aws cloudformation deploy --stack-name grc-capstone-network-
stack --template-file cloudformation-network-stack.yaml --capabilities CAPABILITY_NAMED_IAM --parameter-overrides Envi
ronmentName=grc-capstone --region us-east-1

Waiting for changeset to be created..
Waiting for stack create/update to complete
Successfully created/updated stack - grc-capstone-network-stack
eee_W_5998079@runweb219096:~/GRC208-AWS-Capstone-Project$ aws cloudformation describe-stacks --stack-name grc-capstone
-network-stack --query "Stacks[0].StackStatus" --output text
CREATE_COMPLETE
eee_W_5998079@runweb219096:~/GRC208-AWS-Capstone-Project$
```

```
eee_W_5998079@runweb219096:~/GRC208-AWS-Capstone-Project$ aws cloudformation list-exports --query "Exports[?contains(N
ame, 'grc-capstone')].{Name:Name,Value:Value}" --output table
-----------------------------------------------------------------
|                          ListExports                          |
+-----------------------------------+--------------------------+
|               Name                |           Value          |
+-----------------------------------+--------------------------+
|  grc-capstone-alb-sg-id           |  sg-00813fe5d0f3c2bfe    |
|  grc-capstone-ecs-sg-id           |  sg-0f4b009f6dd27e0af    |
|  grc-capstone-private-subnet-1-id |  subnet-0395bfc87329115fc |
|  grc-capstone-private-subnet-2-id |  subnet-0f4855938ac75f687 |
|  grc-capstone-public-subnet-1-id  |  subnet-0f7223a6c7129c5cf |
|  grc-capstone-public-subnet-2-id  |  subnet-0e25a7a76a39b8089 |
|  grc-capstone-rds-sg-id           |  sg-0b30185d31e4cb393    |
|  grc-capstone-vpc-id              |  vpc-0be92c0ee444e21ec   |
+-----------------------------------+--------------------------+
eee_W_5998079@runweb219096:~/GRC208-AWS-Capstone-Project$
```

Phase 2: Database Infrastructure

```
Waiting for changeset to be created..
Waiting for stack create/update to complete
Successfully created/updated stack - grc-capstone-database-stack
eee_W_5998079@runweb219096:~/GRC208-AWS-Capstone-Project$ aws cloudformation describe-stacks --stack-name grc-capstone-database-stack --query "Stacks[0].StackStatus" --ou
tput text
CREATE_COMPLETE
eee_W_5998079@runweb219096:~/GRC208-AWS-Capstone-Project$ aws cloudformation describe-stacks --stack-name grc-capstone-database-stack --query "Stacks[0].Outputs[].{Key:Ou
tputKey,Value:OutputValue}" --output table
-----------------------------------------------------------------------------------------------------------------------
|                                                    DescribeStacks                                                   |
+--------------------+----------------------------------------------------------------------------------------------+
|        Key         |                                                   Value                                      |
+--------------------+----------------------------------------------------------------------------------------------+
|  ControlsTableArn  |  arn:aws:dynamodb:us-east-1:211125455591:table/grc-capstone-controls-grc-capstone-database-stack     |
|  RiskTableArn      |  arn:aws:dynamodb:us-east-1:211125455591:table/grc-capstone-risk-grc-capstone-database-stack         |
|  ComplianceTableArn|  arn:aws:dynamodb:us-east-1:211125455591:table/grc-capstone-compliance-grc-capstone-database-stack   |
|  DatabaseEndpoint  |  grc-capstone-db.cp22ak24co15.us-east-1.rds.amazonaws.com                                          |
+--------------------+----------------------------------------------------------------------------------------------+
eee_W_5998079@runweb219096:~/GRC208-AWS-Capstone-Project$ aws rds describe-db-instances --db-instance-identifier grc-capstone-db --query "DBInstances[0].{Status:DBInstanc
eStatus,Endpoint:Endpoint.Address,Engine:Engine,InstanceClass:DBInstanceClass}" --output table
-------------------------------------------------------------------------------------------------
|                                          DescribeDBInstances                                  |
+-------------------------------------------------------------+---------+----------------+-------------+
|                          Endpoint                           | Engine  | InstanceClass  |   Status    |
+-------------------------------------------------------------+---------+----------------+-------------+
|  grc-capstone-db.cp22ak24co15.us-east-1.rds.amazonaws.com |  mysql  |  db.t3.micro   |  available  |
+-------------------------------------------------------------+---------+----------------+-------------+
eee_W_5998079@runweb219096:~/GRC208-AWS-Capstone-Project$ aws dynamodb list-tables --query "TableNames[?contains(@, 'grc-capstone')]" --output table
-------------------------------------------------------------
|                       ListTables                          |
+-----------------------------------------------------------+
|  grc-capstone-compliance-grc-capstone-database-stack  |
|  grc-capstone-controls-grc-capstone-database-stack    |
|  grc-capstone-risk-grc-capstone-database-stack        |
+-----------------------------------------------------------+
eee_W_5998079@runweb219096:~/GRC208-AWS-Capstone-Project$ aws secretsmanager create-secret --name grc-capstone/db-credentials --secret-string '{"username":"grcadmin","pas
sword":"MyPassword123"}' --region us-east-1
{
    "ARN": "arn:aws:secretsmanager:us-east-1:211125455591:secret:grc-capstone/db-credentials-VoWHri",
    "Name": "grc-capstone/db-credentials",
    "VersionId": "40f4e206-6cea-4a67-8e1d-2eb93789a71f"
}
eee_W_5998079@runweb219096:~/GRC208-AWS-Capstone-Project$
```


```
Waiting for changeset to be created..
Waiting for stack create/update to complete
Successfully created/updated stack - grc-capstone-database-stack
eee_W_5998079@runweb219096:~/GRC208-AWS-Capstone-Project$ aws cloudformation describe-stacks --stack-name grc-capstone-database-stack --query "Stacks[0].StackStatus" --output text
CREATE_COMPLETE
eee_W_5998079@runweb219096:~/GRC208-AWS-Capstone-Project$ aws cloudformation describe-stacks --stack-name grc-capstone-database-stack --query "Stacks[0].Outputs[].{Key:OutputKey,Value:OutputValue}" --output table
-----------------------------------------------------------------------------------------------------------------------
|                                                    DescribeStacks                                                   |
+--------------------+----------------------------------------------------------------------------------------------+
|        Key         |                                                   Value                                      |
+--------------------+----------------------------------------------------------------------------------------------+
|  ControlsTableArn  |  arn:aws:dynamodb:us-east-1:211125455591:table/grc-capstone-controls-grc-capstone-database-stack     |
|  RiskTableArn      |  arn:aws:dynamodb:us-east-1:211125455591:table/grc-capstone-risk-grc-capstone-database-stack         |
|  ComplianceTableArn|  arn:aws:dynamodb:us-east-1:211125455591:table/grc-capstone-compliance-grc-capstone-database-stack   |
|  DatabaseEndpoint  |  grc-capstone-db.cp22ak24co15.us-east-1.rds.amazonaws.com                                          |
+--------------------+----------------------------------------------------------------------------------------------+
eee_W_5998079@runweb219096:~/GRC208-AWS-Capstone-Project$ aws rds describe-db-instances --db-instance-identifier grc-capstone-db --query "DBInstances[0].{Status:DBInstanceStatus,Endpoint:Endpoint.Address,Engine:Engine,InstanceClass:DBInstanceClass}" --output table
-------------------------------------------------------------------------------------------------
|                                          DescribeDBInstances                                  |
+-------------------------------------------------------------+---------+----------------+-------------+
|                          Endpoint                           | Engine  | InstanceClass  |   Status    |
+-------------------------------------------------------------+---------+----------------+-------------+
|  grc-capstone-db.cp22ak24co15.us-east-1.rds.amazonaws.com |  mysql  |  db.t3.micro   |  available  |
+-------------------------------------------------------------+---------+----------------+-------------+
eee_W_5998079@runweb219096:~/GRC208-AWS-Capstone-Project$ aws dynamodb list-tables --query "TableNames[?contains(@, 'grc-capstone')]" --output table
-------------------------------------------------------------
|                       ListTables                          |
+-----------------------------------------------------------+
|  grc-capstone-compliance-grc-capstone-database-stack  |
|  grc-capstone-controls-grc-capstone-database-stack    |
|  grc-capstone-risk-grc-capstone-database-stack        |
+-----------------------------------------------------------+
eee_W_5998079@runweb219096:~/GRC208-AWS-Capstone-Project$ aws secretsmanager create-secret --name grc-capstone/db-credentials --secret-string '{"username":"grcadmin","password":"MyPassword123"}' --region us-east-1
{
    "ARN": "arn:aws:secretsmanager:us-east-1:211125455591:secret:grc-capstone/db-credentials-VoWHri",
    "Name": "grc-capstone/db-credentials",
    "VersionId": "40f4e206-6cea-4a67-8e1d-2eb93789a71f"
}
eee_W_5998079@runweb219096:~/GRC208-AWS-Capstone-Project$
```

Phase 3: Lambda Functions

```
eee_W_5998079@runweb219096:~/GRC208-AWS-Capstone-Project$ echo "=== GRC208 CAPSTONE - COMPLETE RESOURCE SUMMARY ==="
=== GRC208 CAPSTONE - COMPLETE RESOURCE SUMMARY ===
eee_W_5998079@runweb219096:~/GRC208-AWS-Capstone-Project$ echo "" && echo "CloudFormation Stacks:" && aws cloudformation describe-stacks --stack-name grc-capstone-network
-stack --query "Stacks[0].{Stack:StackName,Status:StackStatus}" --output table

CloudFormation Stacks:
-----------------------------------------------------
|                DescribeStacks                     |
+------------------------------+-------------------+
|            Stack             |      Status       |
+------------------------------+-------------------+
|  grc-capstone-network-stack  |  CREATE_COMPLETE  |
+------------------------------+-------------------+
eee_W_5998079@runweb219096:~/GRC208-AWS-Capstone-Project$ aws cloudformation describe-stacks --stack-name grc-capstone-database-stack --query "Stacks[0].{Stack:StackName,
Status:StackStatus}" --output table
-----------------------------------------------------
|                DescribeStacks                     |
+------------------------------+-------------------+
|            Stack             |      Status       |
+------------------------------+-------------------+
|  grc-capstone-database-stack |  CREATE_COMPLETE  |
+------------------------------+-------------------+
eee_W_5998079@runweb219096:~/GRC208-AWS-Capstone-Project$
```

```
eee_W_5998079@runweb219096:~/GRC208-AWS-Capstone-Project$ echo "" && echo "S3 Buckets:" && aws s3api list-buckets --query "Buckets[?contains(Name, 'grc-capstone')].Name"
--output table

S3 Buckets:
-----------------------------------------------------
|                ListBuckets                          |
+-----------------------------------------------------+
|  grc-capstone-compliance-reports-211125455591       |
|  grc-capstone-evidence-bucket-211125455591          |
+-----------------------------------------------------+
eee_W_5998079@runweb219096:~/GRC208-AWS-Capstone-Project$
```

```
eee_W_5998079@runweb219096:~/GRC208-AWS-Capstone-Project$ echo "" && echo "Lambda Functions:" && aws lambda list-functions --query "Functions[?contains(FunctionName, 'grc
')].{Name:FunctionName,Runtime:Runtime,State:State}" --output table

Lambda Functions:
-----------------------------------------------------
|                ListFunctions                        |
+--------------------------+--------------+---------+
|          Name            |   Runtime    | State   |
+--------------------------+--------------+---------+
|  grc-compliance-monitor  |  python3.11  |  None   |
+--------------------------+--------------+---------+
eee_W_5998079@runweb219096:~/GRC208-AWS-Capstone-Project$
```

```
eee_W_5998079@runweb219096:~/GRC208-AWS-Capstone-Project$ echo "" && echo "RDS Database:" && aws rds describe-db-instances --db-instance-identifier grc-capstone-db --quer
y "DBInstances[0].{Status:DBInstanceStatus,Endpoint:Endpoint.Address}" --output table

RDS Database:
-------------------------------------------------------------------------
|                        DescribeDBInstances                            |
+------------------------------------------------------------+-----------+
|                         Endpoint                           |  Status   |
+------------------------------------------------------------+-----------+
|  grc-capstone-db.cp22ak24co15.us-east-1.rds.amazonaws.com  |  available |
+------------------------------------------------------------+-----------+
eee_W_5998079@runweb219096:~/GRC208-AWS-Capstone-Project$
```

```
eee_W_5998079@runweb219096:~/GRC208-AWS-Capstone-Project$ echo "" && echo "DynamoDB Tables:" && aws dynamodb list-tables --query "TableNames[?contains(@, 'grc-capstone')]
" --output table

DynamoDB Tables:
-------------------------------------------------------
|                     ListTables                      |
+-----------------------------------------------------+
|  grc-capstone-compliance-grc-capstone-database-stack  |
|  grc-capstone-controls-grc-capstone-database-stack    |
|  grc-capstone-risk-grc-capstone-database-stack        |
+-----------------------------------------------------+
eee_W_5998079@runweb219096:~/GRC208-AWS-Capstone-Project$
```

The failed aws config

```
eee_W_5998079@runweb219096:~/GRC208-AWS-Capstone-Project$ aws configservice describe-configuration-recorders --region us-east-1 2>&1 | head -10
{
    "ConfigurationRecorders": []
}
eee_W_5998079@runweb219096:~/GRC208-AWS-Capstone-Project$ aws configservice put-configuration-recorder --configuration-recorder name=default,roleARN=arn:aws:iam::21112545
5591:role/LabRole --region us-east-1 2>&1
eee_W_5998079@runweb219096:~/GRC208-AWS-Capstone-Project$ aws configservice start-configuration-recorder --config-recorder-name default --region us-east-1 2>&1

usage: aws [options] <command> <subcommand> [<subcommand> ...] [parameters]
To see help text, you can run:

  aws help
  aws <command> help
  aws <command> <subcommand> help

aws: error: the following arguments are required: --configuration-recorder-name

eee_W_5998079@runweb219096:~/GRC208-AWS-Capstone-Project$ aws configservice start-configuration-recorder --configuration-recorder-name default --region us-east-1 2>&1

An error occurred (NoAvailableDeliveryChannelException) when calling the StartConfigurationRecorder operation: Delivery channel is not available to start configuration re
corder.
eee_W_5998079@runweb219096:~/GRC208-AWS-Capstone-Project$ aws s3api create-bucket --bucket grc-config-bucket-211125455591 --region us-east-1 2>&1
{
    "Location": "/grc-config-bucket-211125455591"
}
eee_W_5998079@runweb219096:~/GRC208-AWS-Capstone-Project$ aws configservice put-delivery-channel --delivery-channel name=default,s3BucketName=grc-config-bucket-2111254555
91 --region us-east-1 2>&1

An error occurred (InsufficientDeliveryPolicyException) when calling the PutDeliveryChannel operation: Insufficient delivery policy to s3 bucket: grc-config-bucket-211125
455591, unable to assume role: arn:aws:iam::211125455591:role/LabRole.
```

DATABASE INITIALIZATION

```
eee_W_5998079@runweb219096:~/GRC208-AWS-Capstone-Project$ which mysql 2>&1
/usr/bin/mysql
eee_W_5998079@runweb219096:~/GRC208-AWS-Capstone-Project$ curl -s https://raw.githubusercontent.com/icdfa/GRC208-AWS-Capstone-Project/main/sample_data.sql -o sample_data.
sql 2>&1
eee_W_5998079@runweb219096:~/GRC208-AWS-Capstone-Project$ ls -la sample_data.sql 2>&1
-rw-rw-rw- 1 eee_W_5998079 eee_S_3193412 10489 Mar 29 18:24 sample_data.sql
eee_W_5998079@runweb219096:~/GRC208-AWS-Capstone-Project$ aws rds describe-db-instances --db-instance-identifier grc-capstone-db --query "DBInstances[0].Endpoint.Address"
 --output text --region us-east-1 2>&1
grc-capstone-db.cp22ak24co15.us-east-1.rds.amazonaws.com
eee_W_5998079@runweb219096:~/GRC208-AWS-Capstone-Project$ mysql -h grc-capstone-db.cp22ak24co15.us-east-1.rds.amazonaws.com -u grcadmin -pMyPassword123 grcdb < sample_dat
a.sql 2>&1
mysql: [Warning] Using a password on the command line interface can be insecure.

^C
```

Environment Variables

```
eee_W_5998079@runweb219096:~/GRC208-AWS-Capstone-Project$ cat > .env << 'EOF'
> EOF
eee_W_5998079@runweb219096:~/GRC208-AWS-Capstone-Project$ echo "# GRC208 Capstone - Environment Configuration" >> .env
eee_W_5998079@runweb219096:~/GRC208-AWS-Capstone-Project$ echo "AWS_REGION=us-east-1" >> .env
eee_W_5998079@runweb219096:~/GRC208-AWS-Capstone-Project$ echo "AWS_ACCOUNT_ID=211125455591" >> .env
eee_W_5998079@runweb219096:~/GRC208-AWS-Capstone-Project$ echo "DB_HOST=grc-capstone-db.cp22ak24co15.us-east-1.rds.amazonaws.com" >> .env
eee_W_5998079@runweb219096:~/GRC208-AWS-Capstone-Project$ echo "DB_PORT=3306" >> .env
eee_W_5998079@runweb219096:~/GRC208-AWS-Capstone-Project$ echo "DB_NAME=grcdb" >> .env
eee_W_5998079@runweb219096:~/GRC208-AWS-Capstone-Project$ echo "DB_USER=grcadmin" >> .env
eee_W_5998079@runweb219096:~/GRC208-AWS-Capstone-Project$ echo "DB_PASSWORD=MyPassword123" >> .env
eee_W_5998079@runweb219096:~/GRC208-AWS-Capstone-Project$ echo "EVIDENCE_BUCKET=grc-capstone-evidence-bucket-211125455591" >> .env
eee_W_5998079@runweb219096:~/GRC208-AWS-Capstone-Project$ echo "COMPLIANCE_REPORTS_BUCKET=grc-capstone-compliance-reports-211125455591" >> .env
eee_W_5998079@runweb219096:~/GRC208-AWS-Capstone-Project$ echo "LAMBDA_FUNCTION_NAME=grc-compliance-monitor" >> .env
eee_W_5998079@runweb219096:~/GRC208-AWS-Capstone-Project$ echo "LAMBDA_ROLE_ARN=arn:aws:iam::211125455591:role/LabRole" >> .env
eee_W_5998079@runweb219096:~/GRC208-AWS-Capstone-Project$ echo "NETWORK_STACK=grc-capstone-network-stack" >> .env
eee_W_5998079@runweb219096:~/GRC208-AWS-Capstone-Project$ echo "DATABASE_STACK=grc-capstone-database-stack" >> .env
eee_W_5998079@runweb219096:~/GRC208-AWS-Capstone-Project$ cat .env
# GRC208 Capstone - Environment Configuration
AWS_REGION=us-east-1
AWS_ACCOUNT_ID=211125455591
DB_HOST=grc-capstone-db.cp22ak24co15.us-east-1.rds.amazonaws.com
DB_PORT=3306
DB_NAME=grcdb
DB_USER=grcadmin
DB_PASSWORD=MyPassword123
EVIDENCE_BUCKET=grc-capstone-evidence-bucket-211125455591
COMPLIANCE_REPORTS_BUCKET=grc-capstone-compliance-reports-211125455591
LAMBDA_FUNCTION_NAME=grc-compliance-monitor
LAMBDA_ROLE_ARN=arn:aws:iam::211125455591:role/LabRole
NETWORK_STACK=grc-capstone-network-stack
DATABASE_STACK=grc-capstone-database-stack
eee_W_5998079@runweb219096:~/GRC208-AWS-Capstone-Project$
```

CloudWatch Alarms

```
eee_W_5998079@runweb219096:~/GRC208-AWS-Capstone-Project$ aws cloudwatch put-metric-alarm --alarm-name "grc-lambda-errors" --alarm-description "Alert if compliance Lambda
 has errors" --metric-name Errors --namespace AWS/Lambda --statistic Sum --period 300 --threshold 1 --comparison-operator GreaterThanOrEqualToThreshold --evaluation-perio
ds 1 --dimensions Name=FunctionName,Value=grc-compliance-monitor --region us-east-1
eee_W_5998079@runweb219096:~/GRC208-AWS-Capstone-Project$ aws cloudwatch describe-alarms --alarm-names "grc-lambda-errors" --query "MetricAlarms[0].{Name:AlarmName,State:
StateValue,Desc:AlarmDescription}" --output table --region us-east-1
-----------------------------------------------------------------------------------------
|                                   DescribeAlarms                                      |
+-------------------------------------+---------------------+---------------------+
|                Desc                 |        Name         |        State        |
+-------------------------------------+---------------------+---------------------+
|  Alert if compliance Lambda has errors|  grc-lambda-errors  |  INSUFFICIENT_DATA  |
+-------------------------------------+---------------------+---------------------+
eee_W_5998079@runweb219096:~/GRC208-AWS-Capstone-Project$ aws cloudwatch put-metric-alarm --alarm-name "grc-rds-high-cpu" --alarm-description "Alert if RDS CPU exceeds 80
%" --metric-name CPUUtilization --namespace AWS/RDS --statistic Average --period 300 --threshold 80 --comparison-operator GreaterThanThreshold --evaluation-periods 2 --di
mensions Name=DBInstanceIdentifier,Value=grc-capstone-db --region us-east-1
eee_W_5998079@runweb219096:~/GRC208-AWS-Capstone-Project$ aws cloudwatch describe-alarms --query "MetricAlarms[].{Name:AlarmName,State:StateValue}" --output table --regio
n us-east-1
---------------------------------------------
|              DescribeAlarms               |
+--------------------+----------------------+
|        Name        |        State         |
+--------------------+----------------------+
|  grc-lambda-errors |  INSUFFICIENT_DATA   |
|  grc-rds-high-cpu  |  OK                  |
+--------------------+----------------------+
eee_W_5998079@runweb219096:~/GRC208-AWS-Capstone-Project$
```

Run Test Suite

```
eee_W_5998079@runweb219096:~/GRC208-AWS-Capstone-Project$ curl -s https://raw.githubusercontent.com/icdfa/GRC208-AWS-Capstone-Project/main/test_cases.py -o test_cases.py
eee_W_5998079@runweb219096:~/GRC208-AWS-Capstone-Project$ ls -la test_cases.py
-rw-rw-rw- 1 eee_W_5998079 eee_S_3193412 13656 Mar 29 18:46 test_cases.py
eee_W_5998079@runweb219096:~/GRC208-AWS-Capstone-Project$ python3 --version
Python 3.7.5
eee_W_5998079@runweb219096:~/GRC208-AWS-Capstone-Project$ python3 test_cases.py 2>&1 | head -30
test_compliance_percentage_calculation (__main__.TestComplianceMonitoring)
Test compliance percentage calculation ... ok
test_compliance_report_generation (__main__.TestComplianceMonitoring)
Test compliance report generation ... ok
test_non_compliant_rules_detection (__main__.TestComplianceMonitoring)
Test detection of non-compliant rules ... ok
test_risk_level_classification (__main__.TestRiskAssessment)
Test risk level classification based on compliance ... ok
test_risk_matrix_values (__main__.TestRiskAssessment)
Test risk matrix scoring values ... ok
test_risk_score_calculation (__main__.TestRiskAssessment)
Test risk score calculation ... ok
test_asset_classification_validation (__main__.TestDataValidation)
Test asset classification validation ... ok
test_control_id_format_validation (__main__.TestDataValidation)
Test control ID format validation ... ok
test_criticality_level_validation (__main__.TestDataValidation)
Test criticality level validation ... ok
test_risk_status_validation (__main__.TestDataValidation)
Test risk status validation ... ok
test_compliance_snapshot_structure (__main__.TestDatabaseOperations)
Test compliance snapshot data structure ... ok
test_control_record_structure (__main__.TestDatabaseOperations)
Test control record structure ... ok
test_risk_register_structure (__main__.TestDatabaseOperations)
Test risk register data structure ... ok
test_framework_control_count (__main__.TestComplianceFrameworks)
Test framework control count ... ok
test_framework_version_format (__main__.TestComplianceFrameworks)
Test framework version format ... ok
eee_W_5998079@runweb219096:~/GRC208-AWS-Capstone-Project$
```

```
eee_W_5998079@runweb219096:~/GRC208-AWS-Capstone-Project$ python3 test_cases.py 2>&1 | tail -10
Test risk report contains required content ... ok
test_compliance_to_risk_workflow (__main__.TestIntegration)
Test workflow from compliance check to risk assessment ... ok
test_control_to_audit_workflow (__main__.TestIntegration)
Test workflow from control implementation to audit logging ... ok

----------------------------------------------------------------------
Ran 22 tests in 0.001s

OK
eee_W_5998079@runweb219096:~/GRC208-AWS-Capstone-Project$
```

Enable CloudTrail Audit Logging

```
eee_W_5998079@runweb219096:~/GRC208-AWS-Capstone-Project$ aws s3api put-bucket-policy --bucket grc-cloudtrail-logs-211125455591 --policy '{"Version":"2012-10-17","Stateme
nt":[{"Sid":"AWSCloudTrailAclCheck","Effect":"Allow","Principal":{"Service":"cloudtrail.amazonaws.com"},"Action":"s3:GetBucketAcl","Resource":"arn:aws:s3:::grc-cloudtrail
-logs-211125455591"},{"Sid":"AWSCloudTrailWrite","Effect":"Allow","Principal":{"Service":"cloudtrail.amazonaws.com"},"Action":"s3:PutObject","Resource":"arn:aws:s3:::grc-
cloudtrail-logs-211125455591/*","Condition":{"StringEquals":{"s3:x-amz-acl":"bucket-owner-full-control"}}}]}' --region us-east-1
eee_W_5998079@runweb219096:~/GRC208-AWS-Capstone-Project$ aws cloudtrail create-trail --name grc-trail --s3-bucket-name grc-cloudtrail-logs-211125455591 --region us-east-
1
{
    "Name": "grc-trail",
    "S3BucketName": "grc-cloudtrail-logs-211125455591",
    "IncludeGlobalServiceEvents": true,
    "IsMultiRegionTrail": false,
    "TrailARN": "arn:aws:cloudtrail:us-east-1:211125455591:trail/grc-trail",
    "LogFileValidationEnabled": false,
    "IsOrganizationTrail": false
}
eee_W_5998079@runweb219096:~/GRC208-AWS-Capstone-Project$
```

```
eee_W_5998079@runweb219096:~/GRC208-AWS-Capstone-Project$ aws s3api put-bucket-policy --bucket grc-cloudtrail-logs-211125455591 --policy '{"Version":"2012-10-17","Stateme
nt":[{"Sid":"AWSCloudTrailAclCheck","Effect":"Allow","Principal":{"Service":"cloudtrail.amazonaws.com"},"Action":"s3:GetBucketAcl","Resource":"arn:aws:s3:::grc-cloudtrail
-logs-211125455591"},{"Sid":"AWSCloudTrailWrite","Effect":"Allow","Principal":{"Service":"cloudtrail.amazonaws.com"},"Action":"s3:PutObject","Resource":"arn:aws:s3:::grc-
cloudtrail-logs-211125455591/*","Condition":{"StringEquals":{"s3:x-amz-acl":"bucket-owner-full-control"}}}]}' --region us-east-1
eee_W_5998079@runweb219096:~/GRC208-AWS-Capstone-Project$ aws cloudtrail create-trail --name grc-trail --s3-bucket-name grc-cloudtrail-logs-211125455591 --region us-east-
1
{
    "Name": "grc-trail",
    "S3BucketName": "grc-cloudtrail-logs-211125455591",
    "IncludeGlobalServiceEvents": true,
    "IsMultiRegionTrail": false,
    "TrailARN": "arn:aws:cloudtrail:us-east-1:211125455591:trail/grc-trail",
    "LogFileValidationEnabled": false,
    "IsOrganizationTrail": false
}
eee_W_5998079@runweb219096:~/GRC208-AWS-Capstone-Project$ aws cloudtrail start-logging --name grc-trail --region us-east-1
eee_W_5998079@runweb219096:~/GRC208-AWS-Capstone-Project$ aws events put-rule --name grc-compliance-check --schedule-expression "rate(1 hour)" --state ENABLED --region us
-east-1
{
    "RuleArn": "arn:aws:events:us-east-1:211125455591:rule/grc-compliance-check"
}
eee_W_5998079@runweb219096:~/GRC208-AWS-Capstone-Project$ aws events put-targets --rule grc-compliance-check --targets "Id"="1","Arn"="arn:aws:lambda:us-east-1:2111254555
91:function:grc-compliance-monitor" --region us-east-1
{
    "FailedEntryCount": 0,
    "FailedEntries": []
}
eee_W_5998079@runweb219096:~/GRC208-AWS-Capstone-Project$ aws lambda add-permission --function-name grc-compliance-monitor --statement-id grc-eventbridge --action lambda:
InvokeFunction --principal events.amazonaws.com --source-arn arn:aws:events:us-east-1:211125455591:rule/grc-compliance-check --region us-east-1
{
    "Statement": "{\"Sid\":\"grc-eventbridge\",\"Effect\":\"Allow\",\"Principal\":{\"Service\":\"events.amazonaws.com\"},\"Action\":\"lambda:InvokeFunction\",\"Resource\"
:\"arn:aws:lambda:us-east-1:211125455591:function:grc-compliance-monitor\",\"Condition\":{\"ArnLike\":{\"AWS:SourceArn\":\"arn:aws:events:us-east-1:211125455591:rule/grc-
compliance-check\"}}}"
}
eee_W_5998079@runweb219096:~/GRC208-AWS-Capstone-Project$ aws events describe-rule --name grc-compliance-check --region us-east-1
{
    "Name": "grc-compliance-check",
    "Arn": "arn:aws:events:us-east-1:211125455591:rule/grc-compliance-check",
    "ScheduleExpression": "rate(1 hour)",
    "State": "ENABLED",
    "EventBusName": "default",
    "CreatedBy": "211125455591"
}
eee_W_5998079@runweb219096:~/GRC208-AWS-Capstone-Project$
```

Create RDS Snapshot Backup

```
eee_W_5998079@runweb219096:~/GRC208-AWS-Capstone-Project$ aws rds create-db-snapshot --db-instance-identifier grc-capstone-db --db-snapshot-identifier grc-capstone-snapsh
ot-$(date +%Y%m%d) --region us-east-1
{
    "DBSnapshot": {
        "DBSnapshotIdentifier": "grc-capstone-snapshot-20260329",
        "DBInstanceIdentifier": "grc-capstone-db",
        "Engine": "mysql",
        "AllocatedStorage": 20,
        "Status": "creating",
        "Port": 3306,
        "AvailabilityZone": "us-east-1b",
        "VpcId": "vpc-0be92c0ee444e21ec",
        "InstanceCreateTime": "2026-03-30T00:24:49.387000+00:00",
        "MasterUsername": "grcadmin",
        "EngineVersion": "8.0.44",
        "LicenseModel": "general-public-license",
        "SnapshotType": "manual",
        "OptionGroupName": "default:mysql-8-0",
        "PercentProgress": 0,
        "StorageType": "gp2",
        "Encrypted": false,
        "DBSnapshotArn": "arn:aws:rds:us-east-1:211125455591:snapshot:grc-capstone-snapshot-20260329",
        "IAMDatabaseAuthenticationEnabled": false,
        "ProcessorFeatures": [],
        "DbiResourceId": "db-JPGHY6YICRXI3TIGW4HB2SQ4VQ",
        "TagList": []
    }
}
eee_W_5998079@runweb219096:~/GRC208-AWS-Capstone-Project$ aws rds describe-db-snapshots --db-snapshot-identifier grc-capstone-snapshot-20260329 --query "DBSnapshots[0].{I
D:DBSnapshotIdentifier,Status:Status,Created:SnapshotCreateTime}" --output table --region us-east-1
-----------------------------------------------------------------------------------
|                                DescribeDBSnapshots                              |
+----------------------------------+--------------------------------+-----------+
|              Created             |               ID               |  Status   |
+----------------------------------+--------------------------------+-----------+
|  2026-03-30T01:59:11.503000+00:00 |  grc-capstone-snapshot-20260329  |  creating |
+----------------------------------+--------------------------------+-----------+
eee_W_5998079@runweb219096:~/GRC208-AWS-Capstone-Project$
```

```
eee_W_5998079@runweb219600:~$ aws rds describe-db-snapshots --snapshot-type manual --query "DBSnapshots[].{ID:DBSnapshotIdentifier,Status:Status,Instance:DBInstanceIdent
ifier}" --output table --region us-east-1
-----------------------------------------------------------------------
|                        DescribeDBSnapshots                          |
+----------------------------------+-------------------+------------+
|               ID                 |     Instance      |  Status    |
+----------------------------------+-------------------+------------+
|  grc-capstone-snapshot-20260329  |  grc-capstone-db  |  available |
+----------------------------------+-------------------+------------+
eee_W_5998079@runweb219600:~$
```

PROJECT SUMMARY

```
eee_W_5998079@runweb219600:~$ aws s3 ls | grep grc
2026-03-29 17:49:24 grc-capstone-compliance-reports-211125455591
2026-03-29 17:48:54 grc-capstone-evidence-bucket-211125455591
2026-03-29 18:51:45 grc-cloudtrail-logs-211125455591
2026-03-29 18:21:53 grc-config-bucket-211125455591
eee_W_5998079@runweb219600:~$ aws lambda list-functions --query "Functions[?contains(FunctionName,'grc')].{Name:FunctionName,Runtime:Runtime}" --output table --region us
-east-1
-----------------------------------------
|             ListFunctions             |
+-------------------------+-------------+
|          Name           |   Runtime   |
+-------------------------+-------------+
|  grc-compliance-monitor |  python3.11 |
+-------------------------+-------------+
eee_W_5998079@runweb219600:~$ ✅grc-compliance-monitor (Python 3.11)
bash: syntax error near unexpected token `('
eee_W_5998079@runweb219600:~$ aws cloudwatch describe-alarms --query "MetricAlarms[].{Name:AlarmName,State:StateValue}" --output table --region us-east-1
-----------------------------------------
|             DescribeAlarms            |
+--------------------+------------------+
|        Name        |       State      |
+--------------------+------------------+
|  grc-lambda-errors |  INSUFFICIENT_DATA |
|  grc-rds-high-cpu  |  OK              |
+--------------------+------------------+
eee_W_5998079@runweb219600:~$ aws cloudtrail describe-trails --query "trailList[].{Name:Name,S3Bucket:S3BucketName,Logging:IsLogging}" --output table --region us-east-1
-----------------------------------------------------------------
|                      DescribeTrails                           |
+---------+------------+------------------------------------+
| Logging |   Name     |             S3Bucket               |
+---------+------------+------------------------------------+
|  None   |  grc-trail |  grc-cloudtrail-logs-211125455591  |
+---------+------------+------------------------------------+
eee_W_5998079@runweb219600:~$
```

```
eee_W_5998079@runweb219600:~$ aws events describe-rule --name grc-compliance-check --region us-east-1
{
    "Name": "grc-compliance-check",
    "Arn": "arn:aws:events:us-east-1:211125455591:rule/grc-compliance-check",
    "ScheduleExpression": "rate(1 hour)",
    "State": "ENABLED",
    "EventBusName": "default",
    "CreatedBy": "211125455591"
}
eee_W_5998079@runweb219600:~$ aws rds describe-db-instances --db-instance-identifier grc-capstone-db --query "DBInstances[0].{DBInstance:DBInstanceIdentifier,Status:DBIn
stanceStatus,Engine:Engine,Endpoint:Endpoint.Address}" --output table --region us-east-1
-------------------------------------------------------------------------------------------
|                                      DescribeDBInstances                                |
+-----------------+-------------------------------------------------------------+---------+-------------+
|   DBInstance    |                          Endpoint                           | Engine  |   Status    |
+-----------------+-------------------------------------------------------------+---------+-------------+
|  grc-capstone-db|  grc-capstone-db.cp22ak24co15.us-east-1.rds.amazonaws.com   |  mysql  |  available  |
+-----------------+-------------------------------------------------------------+---------+-------------+
eee_W_5998079@runweb219600:~$
```